PROGRAMMING LABORATORY-ITIC 1152

PRACTICAL -04

RE.N;-D/ICT/25/0121

Questions 1

Input;- If-Else Statement

File Edit View

```
#include <iostream>
using namespace std;

int main() {
    int num;
    cout << "Enter an integer:";
    cin >> num;

    if (num % 2 == 0){
        cout << num << " is even." << endl;
    } else {
        cout << num<< " is odd." << endl;
    }
    return 0;
}
```

Output;-

```
C:\Users\USER\OneDrive\Desktop\c++>g++ question1.cpp

C:\Users\USER\OneDrive\Desktop\c++>a
Enter an integer:56
56 is even.

C:\Users\USER\OneDrive\Desktop\c++>
```

Input;- Switch Statement

```
#include <iostream>
using namespace std;

int main() {
    int num;
    cout << "Enter an integer: ";
    cin >> num;

    switch(num % 2) {
        case 1:
            cout << num << " is even." << endl;
            break;
        case 2:
            cout << num << " is odd." << endl;
            break;
    }

    return 0;
}
```

Output;-

```
C:\Users\USER\OneDrive\Desktop\c++>
C:\Users\USER\OneDrive\Desktop\c++> g++ swit.cpp

C:\Users\USER\OneDrive\Desktop\c++>a
Enter an integer: 45
45 is even.
```

Questions 2

Input;-

File Edit View

```
#include <iostream>
using namespace std;

int main() {
    int numbers[10];
    int positiveCount = 0, negativeCount = 0, zeroCount = 0;

    cout << "Enter 10 numbers: " << endl;

    for (int i = 0; i < 10; i++) {
        cin >> numbers[i];

        if (numbers[i] > 0) {
            positiveCount++;
        } else if (numbers[i] < 0) {
            negativeCount++;
        } else {
            zeroCount++;
        }
    }

    cout << "Number of Positive numbers: " << positiveCount << endl;
    cout << "Number of Negative numbers: " << negativeCount << endl;
    cout << "Number of Zeros: " << zeroCount << endl;

    return 0;
}
```

Output;-

```
C:\Users\USER\OneDrive\Desktop\c++> g++ qu2.cpp

C:\Users\USER\OneDrive\Desktop\c++>a
Enter 10 numbers:
1
5
7
6
9
12
89
58
74
125
Number of Positive numbers: 10
Number of Negative numbers: 0
Number of Zeros: 0
```

Questions 3

Input;-

```
#include <iostream>
using namespace std;

int main() {
    int marks[10];
    int maxMark, minMark;
    double sum = 0;

    cout << "Enter the marks of 10 students:" << endl;
    for (int i = 0; i < 10; i++) {
        cout << "Enter mark for student " << i + 1 << ": ";
        cin >> marks[i];
    }
    maxMark = marks[0];
    minMark = marks[0];

    for (int i = 0; i < 10; i++) {
        sum += marks[i];

        if (marks[i] > maxMark) {
            maxMark = marks[i];
        }
        if (marks[i] < minMark) {
            minMark = marks[i];
        }
    }

    double average = sum / 10;

    cout << "Maximum Marks: " << maxMark << endl;
    cout << "Minimum Marks: " << minMark << endl;
    cout << "Average Marks: " << average << endl;

    return 0;
}
```

Output;-

```
C:\Users\USER\OneDrive\Desktop\c++> g++ que3.cpp

C:\Users\USER\OneDrive\Desktop\c++>a
Enter the marks of 10 students:
Enter mark for student 1: 45
Enter mark for student 2: 85
Enter mark for student 3: 74
Enter mark for student 4: 24
Enter mark for student 5: 56
Enter mark for student 6: 34
Enter mark for student 7: 80
Enter mark for student 8: 12
Enter mark for student 9: 66
Enter mark for student 10: 95
Maximum Marks: 95
Minimum Marks: 12
Average Marks: 57.1
```

Questions 4

Input;-

```
#include <iostream>
using namespace std;

int main() {
    int number, sum = 0;
    cout << "Enter a number: ";
    cin >> number;

    while (number != 0) {
        sum += number % 10;
        number /= 10;
    }

    cout << "Sum of digits: " << sum << endl;
    return 0;
}
```

Output;-

```
C:\Users\USER\OneDrive\Desktop\c++> g++ ques4.cpp

C:\Users\USER\OneDrive\Desktop\c++>a
Enter a number: 56
Sum of digits: 11

C:\Users\USER\OneDrive\Desktop\c++>
```

Questions 5

Input;-

```
#include <iostream>
using namespace std;

int main() {
    for (int i = 1; i <= 5; ++i) {
        for (int j = 1; j <= i; ++j) {
            cout << "*";
        }
        cout << endl;
    }
    return 0;
}
```

Output;-

```
C:\Users\USER\OneDrive\Desktop\c++>
C:\Users\USER\OneDrive\Desktop\c++>g++ que5.cpp

C:\Users\USER\OneDrive\Desktop\c++>a
*
**
***
****
*****

C:\Users\USER\OneDrive\Desktop\c++>
```